R22A

14/11/2025

Code No.: 22ITC22N

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. (Biotech) III Sem (Main) Examination November 2025

Data Structures Using Python

Time: 3 Hours **Max Marks: 60**

Note: "Answer **ALL** questions from **Part-A** and **Part-B** (including **internal choices**) at one place and in the same order as in the question paper."

Part - A

(5Q X 2M = 10 Marks)

		M	CO	BT
1	What are NumPy arrays? How are they different from Python lists?	(2)	1	1
2	Why is asymptotic analysis preferred over empirical analysis?	(2)	2	1
3	Define Stack ADT. What are its basic operations?	(2)	3	1
4	What is a Graph? Compare various types of graphs?	(2)	4	1,2
5	Outline components of a hash table?	(2)	5	L2

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Define a class in Python. Explain with an example how objects are created from a class.	(5)	1	1,2
	(b)	Develop a Python program to demonstrate array slicing and reshaping in NumPy.	(5)	1	3
		(OR)			
7	(a)	Analyze different ways to handle missing data in Pandas.	(5)	1	4
	(b)	Compare line plots and scatter plots. Identify examples of when to use each.	(5)	1	2,3
8	(a)	What is big O notation? Illustrate with an example of linear search?	(5)	2	1,2
	(b)	Demonstrate working of Selection Sort with an example. Analyze its time complexity.	(5)	2	2,3
		(OR)			
9	(a)	Demonstrate working of Quick Sort with a suitable example.	(5)	2	2
	(b)	Illustrate an example on Radix Sort algorithm.	(5)	2	2
10	(a)	Develop an algorithm to insert and delete nodes in a Doubly Linked List.	(5)	3	3
	(b)	Develop an algorithm for Push and Pop operations in a stack.	(5)	3	3
		(OR)			
11	(a)	Analyze operations of Enqueue and Dequeue in a queue with an example?	(5)	3	4
	(b)	Examine how queues are used in printer task scheduling.	(5)	3	4
12	(a)	Define Binary Search Trees (BSTs). Examine process of insertion, deletion, and searching in BSTs with suitable diagrams.	(5)	4	1,4
	(b)	What are Graph Traversals? Make use of Depth First Search (DFS) with suitable example?	(5)	4	1,3

(OR)

13	(a)	Construct a Binary Search Tree from the following elements and perform Inorder traversal: 50, 30, 70, 20, 40, 60, 80	(5)	4	3
	(b)	Comapre tree traversal with graph traversal techniques?	(5)	4	4
14	(a)	What is the Knuth-Morris-Pratt (KMP) Algorithm? Explain its working and its time complexity with an example?	(5)	5	1,2
	(b)	Analyze the term 'Load Factor' in hashing. How does it affect the performance of a hash table?	(5)	5	1,4
		(OR)			
15	(a)	Explain the construction and working of the prefix table (LPS array) used in KMP algorithm?	(5)	5	2
	(b)	Compare Separate Chaining and Open Addressing techniques for resolving collisions?	(5)	5	4

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